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Optimal time-domain pulse width modulation for three-phase inverters

Siddharth Tyagi   ; Isaak Mayergoyz

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Siddharth Tyagi^{a)}  and Isaak Mayergoyz^{b)}

AFFILIATIONS

ECE Department, University of Maryland, College Park, Maryland 20740, USA

^{a)} Author to whom correspondence should be addressed: styagi@umd.edu

^{b)} isaak@umd.edu

ABSTRACT

An optimal time-domain technique for pulse-width modulation (PWM) in three-phase inverters is presented. This technique is based on the time-domain per phase analysis of three-phase inverters for linear, balanced output load circuits. Exact analytical solutions for the output phase currents are presented for first, second, and third order circuits for any sequence of rectangular pulses. Next, the role of **symmetries** in the structure of three-phase PWM inverter voltages is discussed. Finally, using the exact analytical solutions for phase-currents as well as symmetry considerations, minimization of output current harmonics is set up as an optimization problem. Sample numerical results are presented that highlight improvements in the performance of three-phase inverters.

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I. INTRODUCTION

The principle of pulse-width modulation (PWM) is to generate voltages that are trains (sequences) of rectangular pulses. The widths of these pulses are properly modulated to suppress lower-order voltage harmonics at the expense of higher order harmonics, which are, in turn, suppressed in output currents and voltages by inductors in the inverter circuits.^{1–3}

Usually, the H-bridge topology shown in Fig. 1 is used in the design of three-phase inverters. There are a number of ways to generate pulse width modulated voltages and currents for the three-phase circuit inverter shown in Fig. 1. Space Vector PWM (SVPWM) is the most commonly used method to generate PWM pulses for such inverters.¹ Over the years, extensive research has been performed on various aspects of PWM.^{4–12}

In this paper, a time-domain per-phase analysis of inverters is performed to derive analytical expressions for phase currents, which are then used for minimization of their harmonic-contents. Furthermore, in the framework of the developed technique, specific lower-order harmonics can be completely eliminated by imposing certain constraints on the minimization problem. In this way, selective harmonic elimination (SHE) is achieved simultaneously with minimization of Total Harmonic Distortion (THD).

This manuscript is organized as follows: In Sec. II, we present a time-domain analysis of PWM for three-phase inverters. The PWM voltages are fully characterized by switching time-instants. The exact analytical solutions for phase-currents for first, second, and third order circuits commonly used in inverters are obtained in terms of these time-instants. Similar analytical expressions can be derived for linear electric circuits of any order, and they can be used for the time-domain analysis for various PWM techniques. By using these analytical solutions, the problem of optimal PWM design can be framed as a minimization problem.

Symmetries play an important role in the formation of PWM line-voltages in three-phase inverters. In Sec. III, we discuss the mathematical and physical aspects of symmetries involved in the performance of PWM inverters. It turns out that these symmetry considerations impose specific constraints on the switching time-instants that describe PWM three-phase line-voltages. These symmetry constraints are very general in nature, and they are valid for any PWM technique. In the case of the optimal time-domain technique, these constraints appreciably reduce the number of unknowns involved in the optimization process.

In Sec. IV, some mathematical details of the optimization technique are discussed, and sample numerical results are presented,

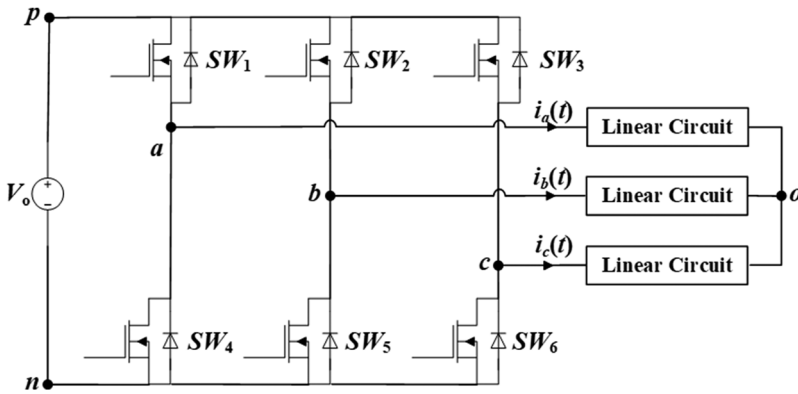


FIG. 1. 3-Phase H-bridge inverter.

which highlight the improvements in the performance of three-phase inverters.

II. TIME-DOMAIN ANALYSIS

A. Per-phase analysis of three-phase inverter

In the three-phase H-bridge inverter shown in Fig. 1, the three-phase star-type loads are modeled by linear circuits. It is assumed that the loads are balanced, i.e., they are identical. Some practically useful linear circuits used to model various applications of three-phase inverters are shown in Figs. 3–5. For instance, the LR -circuit is commonly used to model motors and other inductive loads. Similarly, the L - RC and L - C - LR -circuits are commonly employed as models for uninterruptible power supplies (UPS). However, the analysis presented in this subsection is quite general and does not depend on the exact nature of these linear circuits.

Since the output loads are assumed to be linear, the following equations can be written for the phase currents using Kirchhoff's voltage law (KVL):

$$\dot{Z}[i_a(t)] = v_a(t) - v_o(t), \quad (1)$$

$$\dot{Z}[i_b(t)] = v_b(t) - v_o(t), \quad (2)$$

$$\dot{Z}[i_c(t)] = v_c(t) - v_o(t), \quad (3)$$

where $v_a(t)$, $v_b(t)$, $v_c(t)$, and $v_o(t)$ are potentials of nodes a , b , c , and o , respectively, measured with respect to some reference node, for instance, node n , while $\dot{Z}[\cdot]$ is a linear ordinary differential operator of the load electric circuit.

Next, using Kirchhoff's current law, we find

$$i_a(t) + i_b(t) + i_c(t) = 0. \quad (4)$$

Adding Eqs. (1)–(3), and using formula (4) along with the linearity of $\dot{Z}[\cdot]$ and the fact that $\dot{Z}[0] = 0$, we get the following expression for $v_o(t)$:

$$v_o(t) = \frac{v_a(t) + v_b(t) + v_c(t)}{3}. \quad (5)$$

By substituting Eq. (5) into Eq. (1), we find

$$\begin{aligned} \dot{Z}[i_a(t)] &= v_a(t) - \frac{v_a(t) + v_b(t) + v_c(t)}{3} \\ &= \frac{2}{3}v_a(t) - \frac{1}{3}[v_b(t) + v_c(t)] \\ &= \frac{1}{3}[v_a(t) - v_b(t) + v_a(t) - v_c(t)]. \end{aligned} \quad (6)$$

Thus,

$$\dot{Z}[i_a(t)] = \frac{1}{3}[v_{ab}(t) + v_{ac}(t)]. \quad (7)$$

Similarly, we can derive

$$\dot{Z}[i_b(t)] = \frac{1}{3}[v_{bc}(t) + v_{ba}(t)], \quad (8)$$

$$\dot{Z}[i_c(t)] = \frac{1}{3}[v_{ca}(t) + v_{cb}(t)]. \quad (9)$$

We note here that the right-hand sides of Eqs. (7)–(9) depend on the PWM line-voltages that are generated by the inverter, while the left-hand sides of those equations contain the phase-currents. Thus, Eqs. (7)–(9) can be interpreted as the time-domain per-phase model of the inverter. These equations shall be used to derive the analytical expressions for the phase-currents and optimize the harmonic-performance of the inverter.

It is instructive to highlight the following similarity of the above time-domain per-phase model of the inverter to the frequency domain per-phase analysis of 3-phase AC circuits under balanced operation. Indeed, once we obtain the analytical solution for the phase current $i_a(t)$ from Eq. (7) (as discussed in Sec. II B), the analytical expressions for $i_b(t)$ and $i_c(t)$ can be easily obtained as versions of $i_a(t)$ time-shifted by $\frac{T}{3}$ and $\frac{2T}{3}$, respectively. This is the essence of per-phase analysis, where the solution for currents and voltages in one phase yields the complete information about currents and voltages in other phases by appropriate time-shifts.

It is interesting to point out that the right-hand sides of per-phase equations (7)–(9) are two-level voltages produced by single-level pulse width modulated line-voltages $v_{ab}(t)$, $v_{bc}(t)$, and $v_{ca}(t)$, respectively. This is in clear contrast with single-phase PWM inverters, where the currents are driven by single level line-voltages.

B. Analytical expression for phase currents

For three-phase inverters, the line-voltages $v_{ab}(t)$, $v_{bc}(t)$, and $v_{ca}(t)$ are periodic trains of rectangular pulses. We assume that the line-voltage $v_{ab}(t)$ is as shown in Fig. 2. Here, T is the time-period, and the associated frequency is $\omega = \frac{2\pi}{T}$. The voltage $v_{ab}(t)$ must have half-wave symmetry to eliminate even harmonics. This means that

$$v_{ab}\left(t + \frac{T}{2}\right) = -v_{ab}(t). \quad (10)$$

Thus, $v_{ab}(t)$ can be completely characterized by its values in the interval $0 \leq t \leq \frac{T}{2}$. If the number of pulses in the interval $(0, \frac{T}{2})$ is N , then these rectangular pulses can be described by a sequence of strictly monotonically increasing switching time-instants $t_1, t_2, \dots, t_{2N}$. It is clear that the following formula is valid for $v_{ab}(t)$:

$$v_{ab}(t) = \begin{cases} 0, & \text{if } t_{2j} < t < t_{2j+1}, \\ V_o, & \text{if } t_{2j+1} < t < t_{2j+2}, \end{cases} \quad (11)$$

where $j = 0, 1, 2, \dots, N$, and

$$t_0 = 0, \quad t_{2N+1} = \frac{T}{2}. \quad (12)$$

We shall now proceed to derive the expressions for the phase currents $i_a(t)$, $i_b(t)$, and $i_c(t)$ as functions of the switching time-instants that describe the line-voltages. Let $i_{ab}(t)$ and $i_{ac}(t)$ be solutions to the following equations:

$$\hat{Z}[i_{ab}(t)] = v_{ab}(t), \quad (13)$$

$$\hat{Z}[i_{ac}(t)] = v_{ac}(t). \quad (14)$$

Then, due to the linearity of the operator $\hat{Z}[\cdot]$, the solution $i_a(t)$ to Eq. (7) can be written as

$$i_a(t) = \frac{1}{3} [i_{ab}(t) + i_{ac}(t)]. \quad (15)$$

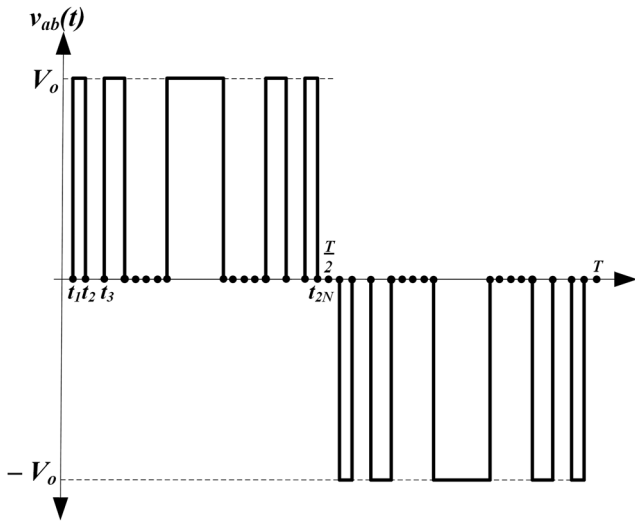


FIG. 2. Structure of the output line-voltage.

Similar expressions can be written for the other phase currents.

It is clear that solutions to Eqs. (13)–(15) depend on the nature of the output circuit. We begin by considering the LR -circuit shown in Fig. 3, for which the linear operator $\hat{Z}[i_{ab}(t)]$ can be written as follows:

$$\hat{Z}[i_{ab}(t)] = L \frac{di_{ab}}{dt}(t) + Ri_{ab}(t). \quad (16)$$

Consecutively, from Eqs. (11), (13), and (16), we can obtain

$$i_{ab}(t) = \begin{cases} A_{2j+1} e^{-\frac{R}{L}t}, & \text{if } t_{2j} < t < t_{2j+1}, \\ A_{2j+2} e^{-\frac{R}{L}t} + \frac{V_o}{R}, & \text{if } t_{2j+1} < t < t_{2j+2}, \end{cases} \quad (17)$$

where the constants A_{2j+1} and A_{2j+2} must be determined by using the continuity of electric current $i_{ab}(t)$ at times t_{2j} and t_{2j+1} as well as the half-wave symmetry boundary condition,

$$i_{ab}\left(\frac{T}{2}\right) = -i_{ab}(0), \quad (18)$$

imposed by the half-wave symmetry [see Eq. (10)] of $v_{ab}(t)$.

From formula (17), using the continuity of $i_{ab}(t)$ at the time-instants $t_1, t_2, \dots, t_{2N}$, as well as the boundary condition (18), we arrive at the following simultaneous equations:

$$A_2 - A_1 = -\frac{V_o}{R} e^{\frac{Rt_1}{L}}, \quad (19)$$

$$A_3 - A_2 = \frac{V_o}{R} e^{\frac{Rt_2}{L}}, \quad (20)$$

⋮

$$A_{2j} - A_{2j-1} = -\frac{V_o}{R} e^{\frac{Rt_{2j-1}}{L}}, \quad (21)$$

$$A_{2j+1} - A_{2j} = \frac{V_o}{R} e^{\frac{Rt_{2j}}{L}}, \quad (22)$$

⋮

$$A_{2N+1} - A_{2N} = \frac{V_o}{R} e^{\frac{Rt_{2N}}{L}}, \quad (23)$$

and

$$A_1 + A_{2N+1} e^{-\frac{RT}{2L}} = 0. \quad (24)$$

These are linear simultaneous equations with a sparse two-diagonal matrix. They can be analytically solved through simple additions to derive the following expressions:

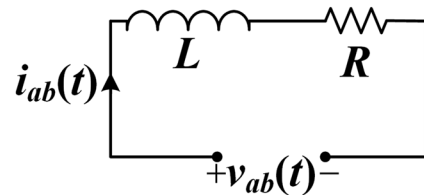


FIG. 3. First order L - R circuit.

$$A_1 = -\frac{V_o}{R} \frac{\sum_{j=1}^{2N} (-1)^j e^{\frac{R}{L} t_j}}{1 + e^{-\frac{RT}{2L}}} e^{-\frac{RT}{2L}} \quad (25)$$

and

$$A_j = A_1 + \frac{V_o}{R} \sum_{n=1}^{j-1} (-1)^n e^{\frac{R}{L} t_n} \quad \text{for } j = 2, 3, \dots, 2N + 1. \quad (26)$$

By using formulas (25) and (26) for the A -coefficients in Eq. (17), we can obtain the general analytical solution for the current $i_{ab}(t, t_1, t_2, \dots, t_{2N})$ in terms of the switching time-instants that describe the voltage $v_{ab}(t)$. Next, we find the analytical expression for $i_{ac}(t)$. We observe that, in order to eliminate all harmonics of orders divisible by three in the line-voltages, the following translational-symmetry condition must be satisfied:

$$v_{ab}(t) = v_{bc}\left(t + \frac{T}{3}\right) = v_{ca}\left(t - \frac{T}{3}\right). \quad (27)$$

Furthermore,

$$v_{ba}(t) = -v_{ab}(t), \quad v_{cb}(t) = -v_{bc}(t), \quad \text{and} \quad v_{ac}(t) = -v_{ca}(t). \quad (28)$$

Equations (27) and (28) imply that $i_{ac}(t)$ is a time-shifted version of $i_{ab}(t)$. Specifically, $i_{ac}(t)$ can be expressed as a function of the switching time-instants $t_1, t_2, \dots, t_{2N}$ as follows:

$$i_{ac}(t, t_1, t_2, \dots, t_{2N}) = -i_{ab}\left(t + \frac{T}{3}, t_1, t_2, \dots, t_{2N}\right). \quad (29)$$

Substituting the analytical expression for $i_{ab}(t)$ given by Eqs. (17), (25), and (26), as well as Eq. (29) in Eq. (15), we arrive at the following expression for $i_a(t)$:

$$i_a(t) = \frac{1}{3} \left[i_{ab}(t, t_1, t_2, \dots, t_{2N}) - i_{ab}\left(t + \frac{T}{3}, t_1, t_2, \dots, t_{2N}\right) \right]. \quad (30)$$

The latter implies that

$$i_a(t) = i_a(t, t_1, t_2, \dots, t_{2N}), \quad (31)$$

which means that the analytical expression for the phase-current $i_a(t)$ in terms of the switching time-instants $t_1, t_2, \dots, t_{2N}$ that describe three-phase line-voltages can be obtained.

We now proceed to the analysis of the L - RC circuit shown in Fig. 4. We begin by solving for $i_{ab}(t)$ in Eq. (13). The following equations can be written for this circuit:

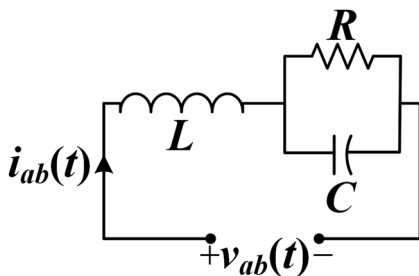


FIG. 4. Second-order L - RC circuit.

$$i_{ab}(t) = C \frac{dv_c}{dt}(t) + \frac{v_c(t)}{R} \quad (32)$$

and

$$LC \frac{d^2 v_c(t)}{dt^2} + \frac{L}{R} \frac{dv_c(t)}{dt} + v_c(t) = v_{ab}(t), \quad (33)$$

where $v_{ab}(t)$ is defined in (11), while $v_c(t)$ is the voltage across the capacitor.

Assuming that the characteristic equation

$$LCs^2 + \frac{L}{R}s + 1 = 0 \quad (34)$$

has distinct roots s_1 and s_2 , the solution to Eq. (33) can be written as

$$v_c(t) = \begin{cases} A_{2j+1}e^{s_1 t} + B_{2j+1}e^{s_2 t}, & \text{if } t_{2j} < t < t_{2j+1}, \\ A_{2j+2}e^{s_1 t} + B_{2j+2}e^{s_2 t} + V_o, & \text{if } t_{2j+1} < t < t_{2j+2}. \end{cases} \quad (35)$$

We intend to obtain the analytical expression for the A_j and B_j coefficients in terms of the switching time-instants. This is done using the continuity of the voltage $v_c(t)$ and its derivative $\frac{dv_c}{dt}(t)$ at the switching time-instants, as well as the half-wave symmetry boundary conditions for $v_c(t)$ and its derivative. This leads to a set of simultaneous equations with a four-diagonal matrix. It turns out that by using a simple mathematical transformation, these equations can be reduced to two decoupled sets of simultaneous equations for the A_j and B_j coefficients, respectively. These are equations with two-diagonal matrices, similar in form to Eqs. (19)–(24). For the A_j coefficients, these equations can be written as follows:

$$A_{j+1} - A_j = (-1)^{j+1} \frac{s_2}{s_1 - s_2} V_o e^{s_1 t_j}, \quad \text{for all } j = 1, 2, \dots, 2N \quad (36)$$

and

$$A_1 + A_{2N+1} e^{\frac{s_1 T}{2}} = 0. \quad (37)$$

Equations (36) and (37) can be solved through simple additions to obtain

$$A_1 = \frac{e^{\frac{s_1 T}{2}}}{1 + e^{\frac{s_1 T}{2}}} \frac{s_2}{s_1 - s_2} V_o \sum_{j=1}^{2N} (-1)^{j+1} e^{s_1 t_j} \quad (38)$$

and

$$A_j = A_1 + \frac{s_2}{s_1 - s_2} V_o \sum_{n=1}^{j-1} (-1)^{n+1} e^{s_1 t_n} \quad \text{for } j = 2, 3, \dots, 2N + 1. \quad (39)$$

Solutions for the B_j coefficients can be obtained by interchanging s_1 and s_2 in Eqs. (38) and (39). Having obtained $v_c(t)$, $i_{ab}(t)$ can be obtained from Eq. (32). Then, using translational symmetry, $i_{ac}(t)$ and consecutively the phase-current $i_a(t)$ can be obtained as analytical functions of the switching time-instants for the L - RC circuit.

Next, we analyze the L - C - LR circuit shown in Fig. 5. This is a third-order circuit (since it has two inductors and a capacitor). By considering the currents $i_{L1}(t)$ and $i_{L2}(t)$ through inductors L_1 and L_2 , respectively, and the voltage $v_c(t)$ across the capacitor C as state-variables, the following state-vector can be introduced:

$$\mathbf{x}(t) = \begin{bmatrix} i_{L1}(t) \\ i_{L2}(t) \\ v_c(t) \end{bmatrix}. \quad (40)$$

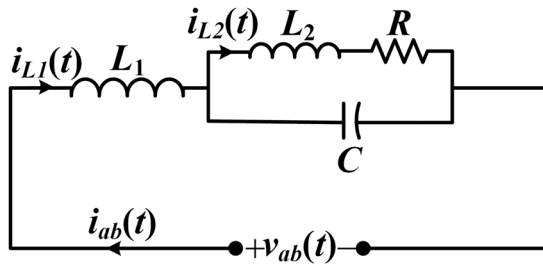


FIG. 5. Third-order L-C-LR circuit.

The following state-space form equations can be easily obtained:

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 0 & 0 & -\frac{1}{L_a} \\ 0 & -\frac{R}{L_b} & \frac{1}{L_b} \\ \frac{1}{C} & -\frac{1}{C} & 0 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} \frac{1}{L_a} \\ 0 \\ 0 \end{bmatrix} v_{ab}(t). \quad (41)$$

Let the eigenvalues s_1 , s_2 , and s_3 of matrix (41) be distinct. Thus, using $v_{ab}(t)$ from Eq. (11), and by noting that $i_{ab}(t) = i_{L1}(t)$, we can write

$$i_{ab}(t) = \begin{cases} A_{2j+1}e^{s_1 t} + B_{2j+1}e^{s_2 t} + C_{2j+1}e^{s_3 t}, & \text{if } t_{2j} < t < t_{2j+1}, \\ A_{2j+2}e^{s_1 t} + B_{2j+2}e^{s_2 t} + C_{2j+2}e^{s_3 t} + \frac{V_o}{R}, & \text{if } t_{2j+1} < t < t_{2j+2}. \end{cases} \quad (42)$$

Again, using the continuity of the current $i_{ab}(t)$ as well as its first and second-order derivatives, along with the half-wave periodic boundary conditions, we can obtain simultaneous equations with a six-diagonal matrix for each switching time-instant. These equations can be reduced to three sets of decoupled equations for A_j , B_j , and C_j coefficients, respectively, with two-diagonal matrices. The solution to the resultant equations for the A_j coefficients is obtained as

$$A_1 = \frac{1}{1 + e^{-s_1 \frac{T}{2}}} \frac{V_o}{R} \frac{s_2 s_3}{(s_1 - s_2)(s_1 - s_3)} \sum_{j=1}^{2N} (-1)^{j+1} e^{-s_1 t_j} \quad (43)$$

and

$$A_j = A_1 - \frac{V_o}{R} \frac{s_2 s_3}{(s_1 - s_2)(s_1 - s_3)} \sum_{n=1}^{j-1} (-1)^{n+1} e^{-s_1 t_n}, \quad (44)$$

for $j = 2, 3, \dots, 2N + 1$.

The formulas for B_j and C_j coefficients have similar forms. Repeating the same steps as for the LR and L-RC cases, the phase current $i_a(t)$ can be obtained. This concludes the analysis of the L-C-LR circuit.

It is worthwhile to point out that similar analytical expressions for phase currents can be derived for linear electric circuits of any order by using the same line of reasoning as above. Furthermore, the derived analytical expressions can be used for time-domain analysis of various PWM techniques. Below, these expressions are utilized to frame the problem of optimal PWM design as a minimization problem.

Before proceeding with the discussion of the optimization problem, we make the following important observation. Equations (27) and (28) imply that all three-phase line-voltages can be described by a single sequence of strictly monotonically increasing switching time-instants $t_1, t_2, \dots, t_{2N}$. However, it turns out that

not any given sequence of strictly monotonically increasing time-instants $t_1, t_2, \dots, t_{2N}$ may, in general, represent three-phase PWM line-voltages. The reason is that time-symmetries of line voltages, as well as the KVL requirement that the voltages $v_{ab}(t)$, $v_{bc}(t)$, and $v_{ca}(t)$ must add up to zero, impose specific constraints on the switching time-instants that describe 3-phase PWM line-voltages. Furthermore, there are also constraints imposed by the fact that only two switches in the same leg of the three-phase inverter in Fig. 1 are usually operated simultaneously. The detailed discussion of these constraints is presented in Sec. III.

C. Time-domain optimization

Now, we shall describe the central idea of the optimal time-domain pulse width modulation technique.

We begin with deriving the expression for the desired fundamental harmonic component of $i_a(t)$. The desired fundamental harmonic components of the line-voltages $v_{ab}(t)$, $v_{bc}(t)$, and $v_{ca}(t)$ can be written as follows:

$$v_{ab,1}(t) = V_m \sin(\omega t), \quad (45)$$

$$v_{bc,1}(t) = V_m \sin\left(\omega t - \frac{2\pi}{3}\right), \quad (46)$$

$$v_{ca,1}(t) = V_m \sin\left(\omega t + \frac{2\pi}{3}\right). \quad (47)$$

In general, for a linear circuit, the desired phase current has the following form:

$$i_{a,1}(t) = I_m \sin(\omega t - \tilde{\phi}), \quad (48)$$

where I_m is the desired peak-value and $\tilde{\phi}$ is the desired phase of the phase current.

As an example, $i_{a,1}(t)$ can be derived for the LR-circuit as follows. Using Eqs. (13), (14), and (16) along with (45) and (47), the desired fundamental harmonic components of currents $i_{ab}(t)$ and $i_{ac}(t)$ can be expressed as follows:

$$i_{ab,1}(t) = \frac{V_m}{\sqrt{R^2 + (\omega L)^2}} \sin(\omega t - \phi), \quad (49)$$

$$i_{ac,1}(t) = -\frac{V_m}{\sqrt{R^2 + (\omega L)^2}} \sin\left(\omega t + \frac{2\pi}{3} - \phi\right), \quad (50)$$

where

$$\tan \phi = \frac{\omega L}{R}. \quad (51)$$

By using Eqs. (49) and (50) as well as formula (15), the fundamental harmonic component of $i_a(t)$ can be obtained. Specifically,

$$i_{a,1}(t) = \frac{V_m}{\sqrt{3}\sqrt{R^2 + (\omega L)^2}} \sin\left(\omega t - \phi - \frac{\pi}{6}\right), \quad (52)$$

which can also be written as

$$i_{a,1}(t) = I_m \sin\left(\omega t - \phi - \frac{\pi}{6}\right), \quad (53)$$

where

$$V_m = \sqrt{3}I_m\sqrt{R^2 + (\omega L)^2}. \quad (54)$$

Similar expressions for I_m and $\tilde{\phi}$ can be obtained for general linear circuits.

Next, we want to find the switching time-instants $t_1, t_2, \dots, t_{2N}$ in Eq. (31) by minimizing in certain sense the difference

$$e(t, t_1, t_2, \dots, t_{2N}) = i_a(t, t_1, t_2, \dots, t_{2N}) - i_{a,1}(t). \quad (55)$$

Specifically, the optimal time-domain pulse width modulation problem can be stated as follows: find such time-instants $t_1, t_2, \dots, t_{2N}$ that the following quantity:

$$E_2(t_1, \dots, t_{2N}) = \int_0^{\frac{T}{2}} \left[i_a(t, t_1, t_2, \dots, t_{2N}) - I_m \sin\left(\omega t - \tilde{\phi}\right) \right]^2 dt \quad (56)$$

reaches its minimum value.

It is apparent that this is the least squares optimization. In mathematical terms, the latter means the optimization of the error-function $i_a(t) - i_{a,1}(t)$ in the L_2 -norm. It is worthwhile to relate the function $E_2(t_1, \dots, t_{2N})$ to the total harmonic distortion (THD) in phase-current $i_a(t)$. The latter is denoted by THD_I , and it is defined as

$$THD_I = \frac{\sqrt{\sum_{n=2}^{\infty} I_n^2}}{I_f}, \quad (57)$$

where I_f is the amplitude of the fundamental harmonic component in $i_a(t)$, while I_n is the amplitude of its n th harmonic. It can be easily verified, by substituting $i_a(t, t_1, \dots, t_{2N})$ in (56) in terms of its Fourier series expansion and using the orthogonality of trigonometric functions,³ that the error integrals $E_2(t_1, \dots, t_{2N})$ and THD_I are related by the following equation:

$$E_2(t_1, t_2, \dots, t_{2N}) = \left[(I_f - I_m)^2 + I_f^2 (THD_I)^2 \right] \cdot \frac{T}{4}. \quad (58)$$

Formula (58) is a special case of the well-known Parseval's equality for the Fourier series. It is evident from formula (58) that the minimization of the function $E_2(t_1, \dots, t_{2N})$ leads to a minimization of the THD in the phase-currents.

It turns out that specific order harmonics in the function $e(t, t_1, \dots, t_{2N})$ defined in (55) can be completely eliminated within the structure of the stated optimization technique. This is done by using constrained optimization. This approach can also be used to ensure that the fundamental harmonic component I_f of the phase-current has the desired value I_m . Specifically, the following constraint can be imposed on the switching time-instants that describe the line-voltage $v_{ab}(t)$:

$$\frac{V_o}{\pi} \sum_{j=1}^{2N} (-1)^{j+1} \cos \omega t_j = V_m, \quad (59)$$

where V_m is defined by formula (54).

Similarly, constraints can be imposed to completely eliminate specific order harmonics. For instance, in order to eliminate the m th harmonic, the following constraint can be used:^{11,12}

$$\sum_{j=1}^{2N} (-1)^j \cos(m\omega t_j) = 0. \quad (60)$$

Thus, the optimization technique can be structured to eliminate specific lower-order harmonics and minimize the total harmonic content of the remaining higher-order harmonics. It is worthwhile to mention that by using the method of Lagrange multipliers, the stated problem can be reduced to unconstrained optimization.

III. CONSTRAINTS ON SWITCHING TIMES IMPOSED BY SYMMETRIES

A. Symmetries

Symmetries play an important role in pulse width modulation of line-voltages in three-phase inverters. Our subsequent discussion deals with the following symmetries:

S1. Translational symmetry: The three-phase line-voltages $v_{ab}(t)$, $v_{bc}(t)$, and $v_{ca}(t)$ are time-shifted versions of each other. Specifically, the following identity is valid:

$$v_{ab}(t) = v_{bc}\left(t + \frac{T}{3}\right) = v_{ca}\left(t - \frac{T}{3}\right). \quad (61)$$

Translational symmetry ensures that the fundamental harmonic components of the line-voltages form a balanced, positive sequence of three-phase voltages.³ Furthermore, it can be shown that translational symmetry results in the elimination of all harmonics of orders divisible by three.

S2. Half-wave symmetry: This symmetry implies that

$$v_{ab}(t) = -v_{ab}\left(t + \frac{T}{2}\right). \quad (62)$$

The same half-wave symmetry is valid for $v_{bc}(t)$ and $v_{ca}(t)$. It can be shown that half-wave symmetry results in the elimination of even-order harmonics in the line-voltages.

S3. Quarter-wave symmetry: The objective of PWM is to generate output voltages that approximate ideal sinusoidal voltages. Hence, it makes intuitive sense to impose the following quarter-wave symmetry condition on the PWM voltages:

$$v_{ab}(t) = v_{ab}\left(\frac{T}{2} - t\right). \quad (63)$$

It is interesting to point out that quarter-wave symmetry (63), half-wave symmetry (62), and periodicity imply that the line-voltage $v_{ab}(t)$ has odd-symmetry. Indeed,

$$v_{ab}(t) = v_{ab}\left(\frac{T}{2} - t\right) = -v_{ab}(T - t) = -v_{ab}(-t). \quad (64)$$

In addition to the above fundamental symmetry conditions, the three-phase line-voltages must satisfy the following constraints:

C1. KVL constraint: The sum of three-phase line-voltages $v_{ab}(t)$, $v_{bc}(t)$, and $v_{ca}(t)$ equals zero,

$$v_{ab}(t) + v_{bc}(t) + v_{ca}(t) = 0. \quad (65)$$

C2. Switching pattern constraint: These are constraints related to the fact that only the states of the two switches in the same leg of the inverter can be simultaneously changed. This prevents unnecessary switchings and helps minimize switching-losses.^{1,9}

B. Role of symmetries in structure of PWM line-voltages

Next, we discuss the implications of the above symmetries and constraints on the structure of the three-phase PWM line-voltages.

The desired fundamental-components of the line-voltages are shown in Fig. 6(a). We begin by dividing the interval $0 \leq t \leq T$ into six equal subintervals of length $\frac{T}{6}$. In each of these subintervals, the PWM pulses of the line-voltages can be grouped together to form a *pulse-group*. Thus, for each line-voltage, each subinterval of length $\frac{T}{6}$ can be characterized by a unique pulse-group.

We first describe the pulse-groups that constitute the PWM line-voltage $v_{ab}(t)$. We label the three pulse-groups in the interval $0 \leq t \leq \frac{T}{2}$ as p^+ , q^+ , and r^+ , respectively. Since $v_{ab,1}(t)$ is positive in the interval $0 \leq t \leq \frac{T}{2}$, $v_{ab}(t)$ shall switch between values 0 and $+V_o$ in this interval, as shown in Fig. 2. Hence, pulse-groups for $v_{ab}(t)$ in the interval $0 \leq t \leq \frac{T}{2}$ are marked by superscript “+.” Furthermore, as a consequence of half-wave symmetry (62), pulses in the interval $\frac{T}{2} \leq t \leq T$ are negative copies of the pulses in $0 \leq t \leq \frac{T}{2}$ (see Fig. 2). Hence, they can be represented by pulse-groups marked using the labels p^- , q^- , and r^- , as shown in Fig. 6(b). The pulses that constitute the pulse-groups p^- , q^- , and r^- have the same widths but opposite polarities as compared to the corresponding pulses in the p^+ , q^+ , and r^+ groups, respectively. Thus, p^+ , q^+ , r^+ , p^- , q^- , and r^- are six distinct pulse-groups that constitute the line-voltage $v_{ab}(t)$. Furthermore, quarter-wave symmetry (63) for $v_{ab}(t)$ implies that the pulses in the p group are mirror images (with respect to $t = \frac{T}{4}$) of those in the r group.

Next, the translational symmetry (61) can be used to determine the pulse-groups in the six subintervals for the line-voltages $v_{bc}(t)$ and $v_{ca}(t)$. Since these line-voltages are time-shifted versions of $v_{ab}(t)$, the pulse-groups in each of the subintervals for $v_{bc}(t)$ and $v_{ca}(t)$ are as shown in Fig. 6(b).

Now, we discuss the implications of the KVL constraint. From Fig. 3(b), we observe that in the time interval $0 \leq t \leq \frac{T}{6}$, the

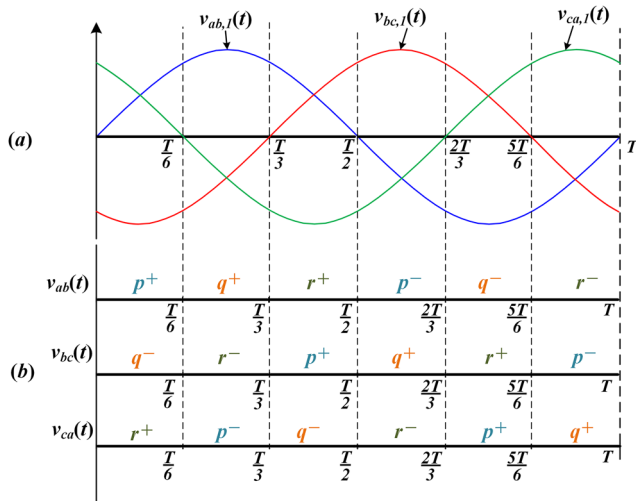


FIG. 6. Structure of 3-phase line-to-line-voltages: (a) desired three-phase sinusoidal fundamental line-voltages and (b) division of the three PWM line-voltages in p , q , and r subgroups.

line-voltages $v_{ab}(t)$, $v_{bc}(t)$, and $v_{ca}(t)$ have pulses of the p^+ , q^- , and r^+ groups, respectively. Similarly, for the subsequent time intervals of length $\frac{T}{6}$, the pulse-groups of these three line-voltages are (q^+, r^-, p^-) , (r^+, p^+, q^-) , (p^-, q^+, r^-) , (q^-, r^+, p^+) , and (r^-, p^-, q^+) , respectively. It is apparent that for each of these time intervals, two of the line-voltages are represented by pulses from the p and r groups of the same sign, while the other line-voltage pulses belong to the q group of the opposite sign. Thus, as a consequence of KVL equation (65) as well as the translational symmetry, for each pulse in the q^+ (or q^-) group, there are corresponding pulses of the opposite polarity in the p^- (or p^+) group and the r^- (or r^+) group such that their total sum is equal to zero. Furthermore, since half-wave symmetry ensures that pulses in the q^+ and q^- groups have the same width but opposite signs, we can arrive at the following important conclusion: half-wave symmetry, translational symmetry, and the KVL constraint imply that each pulse in the q group is the sum of two specific pulses of the same sign: one from the p group and the other from the r group.

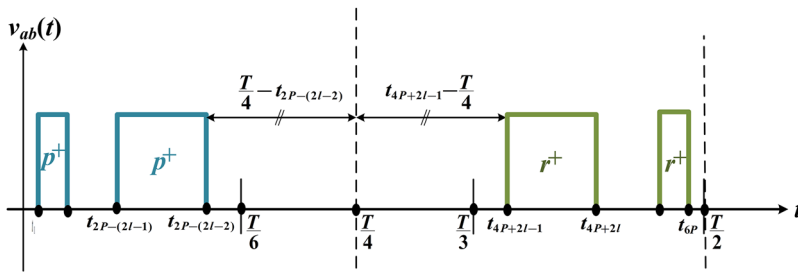
C. Constraints on switching time-instants

We now proceed to discuss the constraints that switching time-instants $t_1, t_2, \dots, t_{2N}$ must satisfy to represent three-phase PWM line-voltages.

First, we determine the number of pulses in three-phase PWM line-voltages. Let the number of pulses in the p group be P . Because of quarter-wave symmetry, pulses in the r group are mirror images of pulses in the p group of the same sign. For this reason, the number of pulses in the r group also equals P . Let the number of pulses in the q group be Q . It is apparent from Fig. 6 and KVL that pulses in the q groups must be wider than the pulses in the p and r groups. Furthermore, it was found in Subsec. III B that pulses in the q group are sums of pulses in the p and r groups of the same sign. This implies that for every pulse in the q group, there must exist one pulse in the p group and one pulse in the r group, which adds to form the given pulse in the q group. This implies that $Q = P$. Thus, we conclude that the number of pulses in the p , q , and r groups is the same and equal to P . This means that $N = 3P = 3Q$. It is desirable that $v_{ab}(t = \frac{T}{4}) = V_o$ [since $v_{ab,1}(t)$ reaches maximum at $\frac{T}{4}$]. For this reason and quarter-wave symmetry, Q is odd. That is, $Q = 2M + 1$, where M is a natural number, and hence $N = 3(2M + 1)$.

Next, we proceed to obtain the algebraic relations that the switching time-instants $t_1, t_2, \dots, t_{6P}$ must satisfy in order to represent three-phase PWM line-voltages. Consider the pulses for line-voltage $v_{ab}(t)$ (see Fig. 2) in the interval $0 \leq t \leq \frac{T}{6}$, that is, pulses in the p^+ group. Each such pulse can be indexed by l , where $l = 0, 1, 2, \dots, P$. The switching time-instants associated with the l th pulse in the p^+ group are t_{2l-1} and t_{2l} . Clearly, time-instants $t_{2P+2l-1}$ and t_{2P+2l} correspond to the l th pulse in the q^+ group, while $t_{4P+2l-1}$ and t_{4P+2l} correspond to the l th pulse in the r^+ group.

Our discussion in Subsection III B suggests that switching time-instants for pulses are the q^+ and r^+ groups that can be obtained from time-instants t_{2l-1} and t_{2l} in the p^+ group. Indeed, since the pulses in r^+ group are mirror images of those in the p^+ group, the corresponding time-instants for pulses in the r^+ group can be obtained from quarter-wave symmetry. Furthermore, as a consequence of half-wave symmetry, translational symmetry, and KVL, each pulse in the q^+ group is the sum of specific pulses in the p^+ and r^+ groups, and hence the time-instants for pulses in the q^+ group can also be

FIG. 7. Relation between time-instants in p and r groups.

obtained in terms of switching time-instants in the p^+ group. We now proceed to derive these relations. The algebraic relations between switching time-instants for pulses in the p^+ and r^+ groups are easily obtained using the quarter-wave symmetry (63), as shown in Fig. 7. As a consequence of quarter-wave symmetry, for every time-instant defining a rising (falling) edge of a pulse in the p^+ group, there is a corresponding time instant defining a falling (rising) edge of a pulse in the r^+ group and these two time-instants are related. Thus, for the l th pulse in the p^+ group, the corresponding time-instants for pulses the r^+ group can be obtained as follows:

$$t_{4P+2l-1} = \frac{T}{2} - t_{2P-(2l-2)}, \quad (66)$$

$$t_{4P+2l} = \frac{T}{2} - t_{2P-(2l-1)}, \quad (67)$$

where $t_{2P-(2l-2)}$ and $t_{2P-(2l-1)}$ are time-instants for pulses in the p^+ group, and $l = 1, 2, \dots, P$.

We now proceed to obtain switching time-instants for pulses in the q^+ group in terms of switching time-instants in the p^+ group. It can be shown that the single-leg switching constraints (C2) lead to two specific patterns on how the KVL constraint (C1) is realized. Specifically, for odd-pulses (i.e., when l is odd), the KVL compensation of the corresponding p , q , and r pulses occurs as shown in Fig. 8. However, for even-pulses (i.e., when l is even), this compensation occurs as shown in Fig. 9. These figures are used below to derive the formulas for switching time-instants for pulses in the q^+ group in terms of switching time-instants for pulses in the p^+ group. Additionally, specific constraints on the switching time-instants for pulses in the p^+ group are established. We now proceed to obtain

switching time-instants for pulses in the q^+ group in terms of switching time-instants in the p^+ group. It can be shown that the single-leg switching constraints (C2) lead to two specific patterns on how the KVL constraint (C1) is realized. Specifically, for odd-pulses (i.e., when l is odd), the KVL compensation of the corresponding p , q , and r pulses occurs as shown in Fig. 8. However, for even-pulses (i.e., when l is even), this compensation occurs as shown in Fig. 9. These figures are used below to derive the formulas for switching time-instants for pulses in the q^+ group in terms of switching time-instants for pulses in the p^+ group. Additionally, specific constraints on the switching time-instants for pulses in the p^+ group are established.

When l is odd (see Fig. 8), the rising-edge of the pulse in the p^+ group corresponds to the rising-edge of the pulse in the q^+ group, the falling-edge of the pulse in the p^+ group corresponds to the rising-edge of the pulse in the r^+ group, while falling-edges of the pulses in the q^+ and r^+ groups are related. Thus, the time-instant $t_{2P+2l-1}$ in the q^+ group is related to t_{2l-1} in the p^+ group as follows (see Fig. 8):

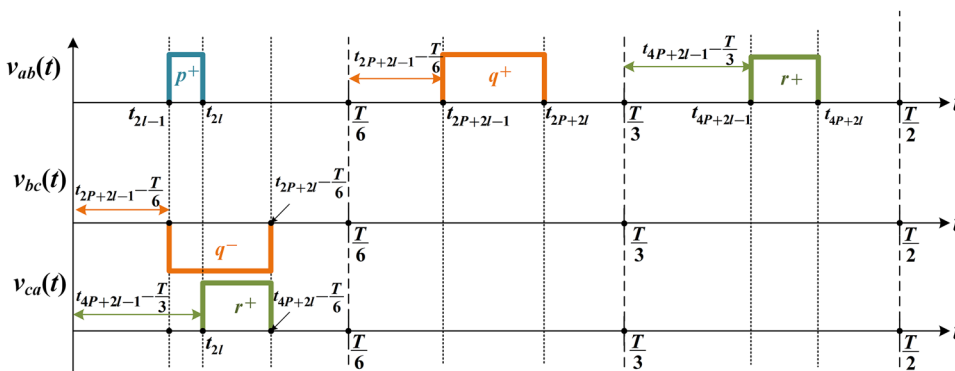
$$t_{2l-1} = t_{2P+2l-1} - \frac{T}{6}, \quad (68)$$

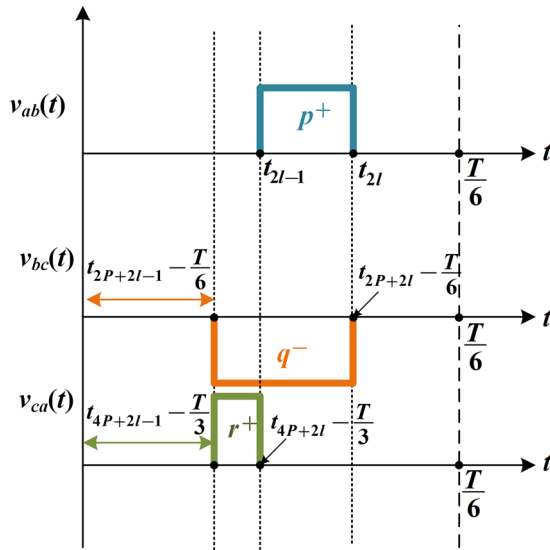
which leads to

$$t_{2P+2l-1} = t_{2l-1} + \frac{T}{6}, \text{ when } l \text{ is odd.} \quad (69)$$

Similarly, the switching time-instant t_{2P+2l} in the q^+ group is related to t_{4P+2l} in the r^+ group as

$$t_{2P+2l} - \frac{T}{6} = t_{4P+2l} - \frac{T}{3}. \quad (70)$$

FIG. 8. Structure of pulses when $l = \text{odd}$.

FIG. 9. Structure of pulses when l is even.

However, using formula (67), we can replace t_{4P+2l} by $\frac{T}{2} - t_{2P-(2l-1)}$, where $t_{2P-(2l-1)}$ is a time-instant in the p^+ group. Thus, Eq. (70) is reduced to

$$t_{2P+2l} = \frac{T}{3} - t_{2P-(2l-1)}, \text{ when } l \text{ is odd.} \quad (71)$$

Equations (69) and (71) relate time-instants in the q^+ group to time-instants in the p^+ group when l is odd.

From Fig. 8, it is also clear that when l is odd, time t_{2l} in the p^+ group and $t_{4P+2l-1}$ in the r^+ group are related. Indeed, from Fig. 8 and using formula (66), we can derive

$$t_{2l} = t_{4P+2l-1} - \frac{T}{3} = \frac{T}{6} - t_{2P-(2l-2)}, \quad (72)$$

which leads to

$$t_{2l} + t_{2P-(2l-2)} = \frac{T}{6}, \text{ when } l \text{ is odd.} \quad (73)$$

Interestingly, both the switching time-instants t_{2l} and $t_{2P-(2l-2)}$ in Eq. (73) belong to the p^+ group. This reveals that not all switching time-instants in the p^+ group are completely independent. Instead, there exist among them mutual algebraic relations of form (73). A similar relation also holds when l is even. This means that the number of independent variables involved in the optimization of the function E_2 needs to be performed is considerably smaller than $2N = 6P$.

Proceeding in the same way as before, the following equations can be derived when l is even by using Fig. 9:

$$t_{2P+2l-1} = \frac{T}{3} - t_{2P-(2l-2)}, \quad (74)$$

$$t_{2P+2l} = t_{2l} + \frac{T}{6}, \quad (75)$$

$$t_{2l-1} + t_{2P-(2l-1)} = \frac{T}{6}. \quad (76)$$

To summarize, we have established that Eqs. (66), (67), (69), (71), and (73)–(76) specify the algebraic relations that the switching time-instants $t_1, t_2, \dots, t_{2P}, t_{2P+1}, \dots, t_{4P}$, and $t_{4P+1}, \dots, t_{6P}$ must satisfy to represent three-phase PWM line-voltages with symmetries (S1)–(S3) under the constraints (C1) and (C2). Imposing these relations as equality constraints on the optimization, a symmetry-preserving time-domain PWM optimization technique can be developed. This matter is further discussed in Sec. IV.

IV. SYMMETRY-PRESERVING OPTIMAL PWM AND NUMERICAL RESULTS

In Sec. I, we defined the function E_2 in Eq. (56) and expressed it as a function of switching time-instants $t_1, t_2, \dots, t_{2N}$. We discussed how minimizing of E_2 leads to the minimization of the harmonic content of the PWM output current [see Eq. (58)]. In Sec. III, we established that $N = 3P$, where P is the number of pulses in each of the p^+ , q^+ , and r^+ groups. Using the notation introduced in Secs. II and III, we can write the objective function as $E_2(t_1, \dots, t_{2P}, t_{2P+1}, \dots, t_{4P}, t_{4P+1}, \dots, t_{6P})$.

We also established that switching time-instants in the q^+ and r^+ groups can be obtained from switching time-instants in the p^+ group, using Eqs. (66), (67), (69), (71), (74), and (75). Moreover, Eqs. (73) and (76) reveal that some switching time-instants in the p^+ group are also related. It can be shown that, when P is odd, there are only $\frac{3P-1}{2}$ independent time-instants, in the sense that with a knowledge of these $\frac{3P-1}{2}$ time-instants, all the $6P$ time-instants can be completely determined using Eqs. (66)–(76). This dramatic reduction in the number of independent variables over which the optimization is performed (from $6P$ to $\frac{3P-1}{2}$) greatly simplifies the numerical computation of the optimization problem.

It is apparent that the time-instants $t_1, t_2, \dots, t_{6P}$ must be strictly monotonically increasing. This constraint can be expressed as the following (non-strict) inequality constraints:

$$t_{j+1} \geq t_j + \tau > t_j, \text{ for all } j = 1, 2, \dots, 6P - 1 \quad (77)$$

and

$$\frac{T}{2} \geq t_{6P} + \tau > t_{6P}, \text{ where } \tau > 0. \quad (78)$$

The inequality constraints in (77) and (78) are used to numerically implement^{13–15} the strict monotonicity condition since most numerical optimization solvers do not accept strict inequalities as inputs. It is worthwhile to mention that, if we define

$$\Delta t_i = t_i - t_{i-1}, \text{ for all } i = 1, 2, 3, \dots, 6P + 1, \quad (79)$$

then, the strict monotonicity constraint can be expressed as

$$\Delta t_i > 0, \text{ for all } i = 1, 2, 3, \dots, 6P + 1. \quad (80)$$

The latter inequalities define a convex region (cone).¹³ For this reason, it may be advantageous to use the variable Δt_i for numerical minimization.

We proceed to implement the aforementioned optimization for the LR-circuit. It is apparent that for this case, the optimal PWM

depends on the parameters R , L , and T . It turns out that this dependence can be expressed in terms of a function of only *one* dimensionless parameter. Indeed, this can be accomplished by introducing the following dimensionless parameter:

$$\alpha = \frac{RT}{L}, \quad (81)$$

and using the scaled-time,

$$\beta = \frac{t}{T} \quad \left(\beta_j = \frac{t_j}{T}, \text{ for all } j = 1, 2, \dots, 2N \right) \quad (82)$$

as well as voltages,

$$v_{R,ab}(t) = Ri_{ab}(t), \quad (83)$$

$$\tilde{v}_{R,ab}(\beta) = v_{R,ab}(\beta T), \quad (84)$$

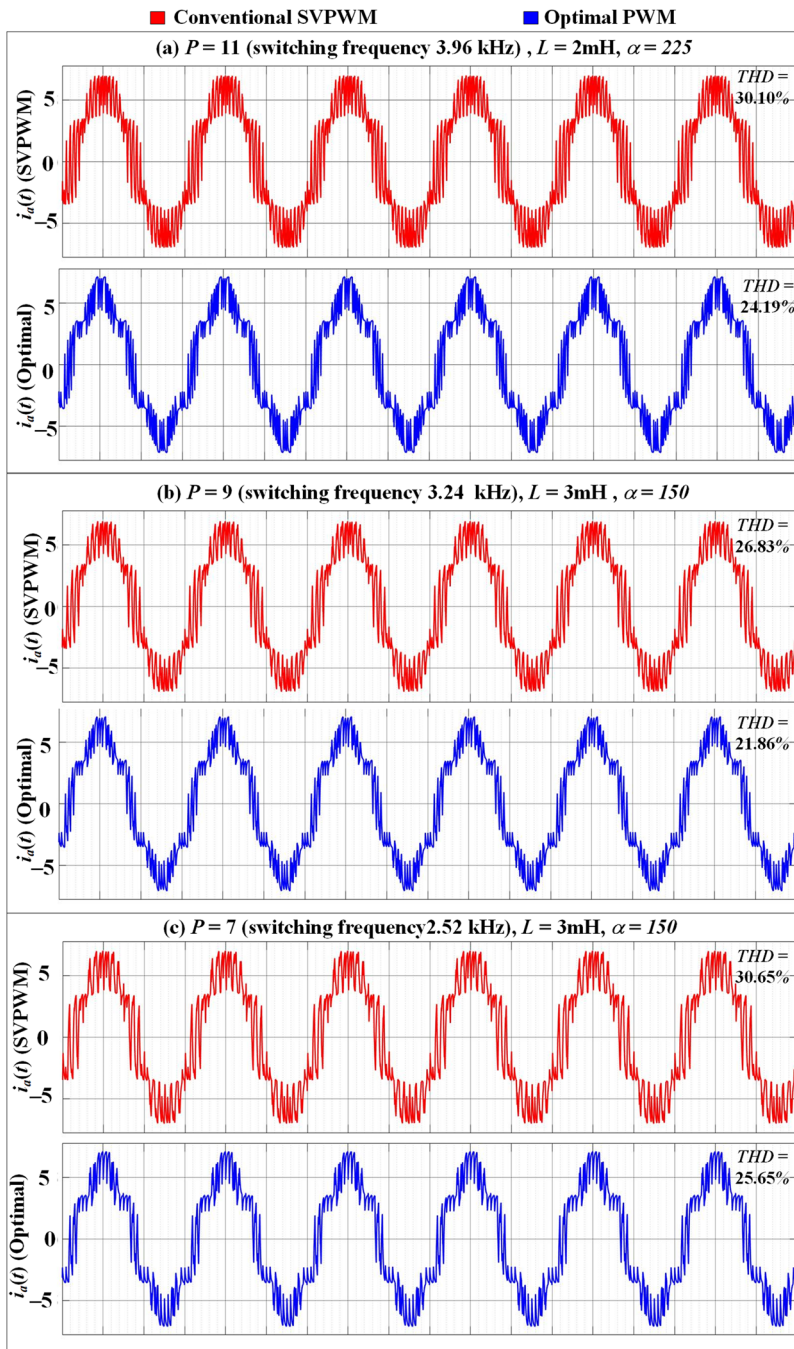


FIG. 10. Comparison of conventional SVPWM with optimal PWM in that case of LR -circuit for $I_m = 5$ A and $R = 27 \Omega$ and different values of P and L .

$$v_{R,a}(t) = Ri_a(t), \quad (85)$$

$$\tilde{v}_{R,a}(\beta) = v_{R,a}(\beta T), \quad (86)$$

and coefficients

$$B_j = RA_j, \quad \text{for all } j = 1, 2, \dots, 2N + 1. \quad (87)$$

Now, Eqs. (17), (25), and (26) can be rewritten as follows:

$$\tilde{v}_{R,ab}(\beta) = \begin{cases} B_{2j+1}e^{-\alpha\beta}, & \text{if } \beta_{2j} < \beta < \beta_{2j+1}, \\ B_{2j+2}e^{-\alpha\beta} + V_o, & \text{if } \beta_{2j+1} < \beta < \beta_{2j+2}, \end{cases} \quad (88)$$

where

$$B_1 = -V_o \frac{\sum_{j=1}^{2N} (-1)^j e^{\alpha\beta_j}}{1 + e^{-\frac{\alpha}{2}}} e^{-\frac{\alpha}{2}} \quad (89)$$

and

$$B_j = B_1 + V_o \sum_{n=1}^{j-1} (-1)^n e^{\alpha\beta_n}, \quad \text{for } j = 2, 3, \dots, 2N + 1. \quad (90)$$

It is evident that (since $N = 3P$)

$$\tilde{v}_{R,ab}(\beta) = \tilde{v}_{R,ab}(\beta, \beta_1, \beta_2, \dots, \beta_{6P}). \quad (91)$$

Thus, using formula (85) along with Eqs. (30), (83), and (84), we obtain

$$\tilde{v}_{R,a}(\beta, \beta_1, \beta_2, \dots, \beta_{6P}) = \frac{1}{3} \left[\tilde{v}_{R,ab}(\beta, \beta_1, \beta_2, \dots, \beta_{6P}) - \tilde{v}_{R,ab}\left(\beta + \frac{1}{3}, \beta_1, \beta_2, \dots, \beta_{6P}\right) \right]. \quad (92)$$

It is clear from Eq. (92) that $\tilde{v}_{R,a}(\beta, \beta_1, \beta_2, \dots, \beta_{6P})$ depends only on the parameter α .

Similarly, the desired fundamental component of the output voltage $v_{R,a}(t)$ can be expressed as

$$\tilde{v}_{R,a,1}(\beta) = v_{R,a,1}(\beta T) = V_{R,m} \sin\left(2\pi\beta - \phi - \frac{\pi}{6}\right), \quad (93)$$

where

$$V_{R,m} = \frac{V_m}{\sqrt{3}\sqrt{1 + \frac{4\pi^2}{\alpha^2}}} \quad (94)$$

and

$$\tan \phi = \frac{2\pi}{\alpha}. \quad (95)$$

If we define the function

$$\tilde{E}_2(\beta_1, \dots, \beta_{6P}) = \int_0^{\frac{1}{2}} \left[\tilde{v}_{R,a}(\beta, \beta_1, \beta_2, \dots, \beta_{6P}) - V_{R,m} \sin\left(2\pi\beta - \phi - \frac{\pi}{6}\right) \right]^2 d\beta, \quad (96)$$

it can be easily verified that

$$\tilde{E}_2(\beta_1, \dots, \beta_{6P}) = \frac{R^2}{T} E_2(t_1, \dots, t_{6P}). \quad (97)$$

Thus, using the above time-scaling, the effects of the parameters R , L , and T on the solutions to the optimization problem can be accounted for by using only the parameter α . Thus, the current-harmonics optimization problem for the LR -circuit can be restated in the standard form^{13–15} as follows.

Minimize the function $\tilde{E}_2(\beta_1, \beta_2, \dots)$ as defined in (97), subject to the strict-monotonicity constraints defined in (77) and (78), as well as constraints (59) and (60) expressed in terms of the β_j variables.

This is the standard problem for constrained non-linear optimization that can be numerically solved using techniques such as interior point methods and sequential quadratic programming.^{14,15}

Below, some sample calculations performed by using the mentioned techniques are presented. These calculations have been performed in MATLAB. Interior-point method and sequential quadratic programming methods have been used for optimization. In these calculations, the value of the bus voltage V_o has been taken to be 300 V and the desired frequency has been chosen to be 60 Hz. The optimization has been performed for different values of inductance L and load resistance R (i.e., for different values of α) as well as for various numbers of pulses P .

Note that P is related to the switching frequency f_{sw} via the following relation:

$$f_{sw} = 6Pf, \quad (98)$$

where $f = \frac{\omega}{2\pi}$. The initial guess for the switching time-instants has been computed according to the conventional Space Vector PWM (SVPWM). The comparative results of the performed calculations are presented in Fig. 10 and Table I.

Next, we report the computational results on PWM optimization with the elimination of specific lower order harmonics. A major advantage of the time-domain technique is that once the switching time-instants defining PWM voltages are known, exact amplitudes

TABLE I. Improvement in THD after optimization when $I_m = 5$ A and $R = 27 \Omega$.

P	f_{sw} (in kHz)	L (in mH)	THD (in %)		
			SVPWM	Optimal PWM	% improvement
5	1.80	5	28.04	23.52	16.12%
7	2.52	3	30.65	25.65	16.17%
9	3.24	3	26.83	21.86	18.52%
11	3.96	2	30.10	24.19	19.63%

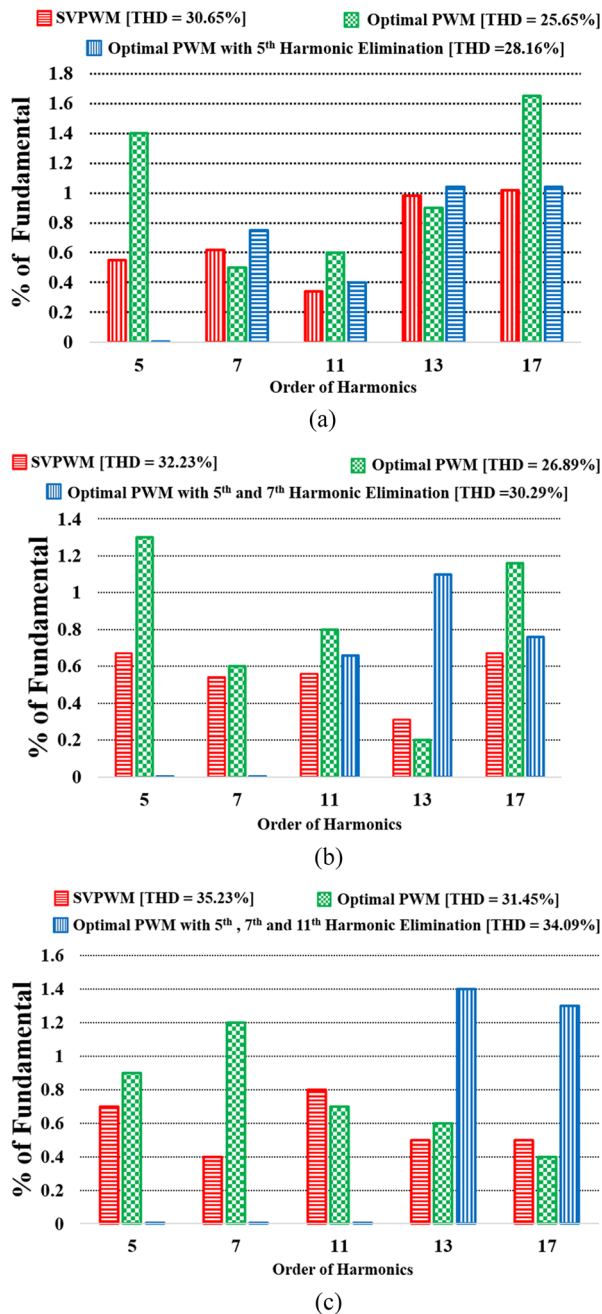


FIG. 11. Computed lower order harmonics for SVPWM, optimal PWM, and optimization with the elimination of (a) 5th order harmonic ($P = 7$, $L = 3$ mH, and $\alpha = 150$), (b) 5th and 7th order harmonics ($P = 9$, $L = 1$ mH, and $\alpha = 50$), and (c) 5th, 7th, and 11th order harmonics ($P = 7$, $L = 1$ mH, and $\alpha = 50$) for $I_m = 5$ A and $R = 27 \Omega$. Computed THD values are also reported.

of lower-order harmonics can be computed without any approximation. Some of these computations for conventional SVPWM and optimized PWM are shown in Fig. 11. From this figure, it can be seen that in some cases, optimization of the total harmonic content

of the PWM current may result in a slight increase in the percentage of lower-order harmonics. This can be resolved by imposing additional nonlinear constraints of the form (60) on the optimization such that specific lower-order harmonics can be eliminated. It is apparent from this figure that imposing SHE constraints yields sub-optimal performance, as far as THD in the current is concerned. However, even after imposing SHE constraints, better performance than conventional SVPWM, is still achieved in terms of THD.

V. CONCLUSION

A per-phase analysis of three-phase inverters is developed for balanced linear circuits connected to the output of the inverters. Time-domain analytical expressions are derived for the phase-currents in terms of switching time-instants that describe three-phase PWM voltages for the LR , L -RC, and L -C- LR circuits. Using these analytical expressions, minimization of harmonics in the output currents and voltages is posed as a standard optimization problem. The use of constrained optimization is proposed for selective harmonic elimination. Furthermore, it is demonstrated that three-phase voltage symmetries, KVL, and switching patterns impose specific algebraic constraints on switching time-instants of three-phase PWM voltages. This leads to a significant reduction in the number of independent variables over which the optimization is performed. It is worthwhile to stress that the obtained symmetry constraints on switching time-instants of three-phase PWM voltages are of a general nature. These constraints can be essential in the design of different PWM techniques. In Sec. IV, it is demonstrated that, for the LR circuit, the dependence of the optimized PWM on parameters R , L , and T can be expressed in terms of a function of only one dimensionless parameter α by appropriate time-scaling. The numerical results revealing improvements in the harmonic performance of inverters using the optimal time-domain optimization technique are presented. The impact of the optimization on lower-order harmonics is analyzed, and elimination of specific lower-order harmonics using constrained optimization is demonstrated.

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